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Overview and application of the AQEM assessment system

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Abstract

The main objective of the European Union (EU) funded project AQEM¹ was to develop a framework of an assessment system for streams in Europe based on benthic macroinvertebrates that fulfils the requirements of the EU Water Framework Directive. Initial assessment methods for 28 European stream types and more generally applicable tools for stream biomonitoring in Europe were generated.

The development of the system was based on a newly collected data set covering stream types in Austria, the Czech Republic, Germany, Greece, Italy, The Netherlands, Portugal and Sweden. Altogether, 901 benthic invertebrate samples were taken using a standardised multi-habitat sampling procedure and a large number of parameters describing the streams and their catchments was recorded for all sampling sites. From the stream and catchment characteristics measures of stress were derived. A large number of metrics was tested independently for each of the stream types, to identify the response of each metric to degradation of a site. This process resulted in up to 18 core metrics for the individual stream types, which were combined into a different multimetric index in each country. The multimetric AQEM assessment system is used to classify a stream stretch into an Ecological Quality Class ranging from 5 (high quality) to 1 (bad quality) and often provides information on the possible causes of degradation.

AQEM provides a taxa list of 9557 European macroinvertebrate taxa with associated autecological information, a software package for performing all the calculations necessary for applying the multimetric AQEM assessment system and a manual describing all aspects of the application of the system from site selection to data interpretation.

Introduction

The EU Water Framework Directive (WFD; Directive 2000/60/EC – Establishing a

Framework for Community Action in the Field of Water Policy) outlines a legal structure for the assessment of all types of water bodies in Europe. A main focus of the WFD is the use of biotic indicators

(macrobenthic fauna, fish fauna and aquatic flora) in stream assessment, which is a concept new to many European countries. Furthermore, the ecological status of a water body is defined by comparing the biological community composition present with the near-natural reference conditions. These new WFD guidelines on ecological water quality assessment have generated a strong demand for either ‘new’ assessment systems fulfilling the criteria or to adapt existing systems to meet the criteria of the WFD. Benthic macroinvertebrates, together with algae, serve as the most common

¹The Development and Testing of an Integrated Assessment System for the Ecological Quality of Streams and Rivers throughout Europe using Benthic Macroinvertebrates.

organism groups used for assessing the ecological river quality (Hellawell, 1986; De Pauw & Hawkes, 1993; Rosenberg & Resh, 1993). Generally, benthic macroinvertebrates are capable of reflecting different anthropogenic perturbations through changes in structure or function in the assemblages and thus enable an overall assessment of streams. Besides organic pollution, which can be assessed using a large number of biological indices, benthic macroinvertebrates can also be used to detect acid stress, habitat loss and overall stream degradation. For these reasons benthic macroinvertebrates are likely to play a major role in future stream assessment in coherence with the Water Framework Directive. The individual European countries have very different traditions in the use of benthic macroinvertebrates for biomonitoring purposes. Overviews were given by Knoben et al. (1995), Nixon et al. (1996) and Birk & Hering (2002). The differences in the methods are inherent in the intensity of sampling and sorting, the taxonomic resolution, the storing and the statistical treatment of the data, the calculation methods used for assessment and the quality assurance procedures used in this process. Therefore, comparisons of results obtained with different national methods are complex and in many cases hindered by the differences. While existing methods in some of the countries are partly fulfilling the requirements of the Water Framework Directive (UK: Wright et al., 2000; France: Agences de l'Eau, 2000), existing methods need adaptation in other countries (Germany: DEV, 1992; Austria: Austrian Standards M 6232, 1997; Chovanec et al., 2000; Netherlands: Peeters et al., 1994; Italy: Ghetti, 1997) or no official method is available (Greece, Portugal).

The central objective of the EU funded project AQEM, which was carried out from 2000 to 2002, aimed at developing a framework for assessing streams in Europe with benthic macroinvertebrates, thus contributing to fulfilling the requirements of the EU Water Framework Directive. From the AQEM mandate, scientifically valid assessment methods were developed for 28 European stream types. The general approach and first results of AQEM have already been published (Buffagni et al., 2001; Feld et al., 2002; Hering et al., 2003; Lorenz et al., 2002; Ofenböck et al., 2002; Pauls et al., 2002; Sommerhäuser & Hering, 2001). The details of the assessment system and further evaluations of the AQEM data set are presented in this special issue.

The specific objective of this article is to give an overview of the process that led to the AQEM as-

essment system. In this paper, we address aspects of AQEM, such as:

- the AQEM approach of assessing the ecological status of running waters
- the sampling design
- the field- and lab methods
- the methods used to identify suitable assessment procedures
- the multimetric assessment system developed in AQEM
- the tools (manual, databases and software) for application of the AQEM assessment system

Method development

Overview

The development of the AQEM assessment system was based on a newly collected data set that covered both the benthic macroinvertebrate fauna and the general stream characteristics. The data were collected by eight countries (Austria, the Czech Republic, Germany, Greece, Italy, The Netherlands, Portugal and Sweden). Generally, the following steps were taken:

- Selection of 29 European stream types within 11 ecoregions (two to five stream types per partner; Table 1), for 28 of which an assessment method was finally developed.
- Focus on the most important stressors (e.g. organic pollution, acidification, degradation in stream morphology or combinations).
- Selection of 11 to 30 sampling sites per stream type covering a biological condition gradient ranging from reference sites (compare criteria in Hering et al., 2003) to heavily degraded conditions.
- Sampling of benthic macroinvertebrates in at least two different seasons using a standardised sampling method. Identification of the macroinvertebrates to the best achievable level (in most cases species level).
- Measuring environmental parameters related to physical/chemical conditions, stream morphology and stream and catchment characteristics (compare Hering et al., 2003).
- Deriving a stream-type specific classification, which reflects the degradation of a site, based either on abiotic data recorded in a harmonised 'site protocol' or on the biotic composition.

- Testing of various attributes of the assemblage (i.e., metrics) with the goal to identify those most affective in measuring the degradation of the stream.
- Selecting those metrics that most strongly correlate with the site's state of degradation measured by chemical or hydromorphological parameters.
- Aggregating the core metrics into a multimetric index.
- Calibrating the stream-type specific assessment systems with independent data.
- Defining condition classes of 'high', 'good', 'moderate', 'poor' and 'bad' ecological status for the selected stream types.

Selection and definition of stream types and stressors

In order to develop assessment systems, as required by the EU Water Framework Directive, stream typologies based on near-natural reference conditions are a crucial pre-requisite. The near-natural reference condition sets the endpoint for the scale of different states of degradation. This measurement scale must be developed for each stream type separately. For example, it is not surprising that the abiotic conditions of slow-flowing lowland streams with finer bed sediments and higher temperatures support completely different bio-coenoses than fast flowing mountain streams with coarse substrata and generally lower water temperatures. Also the effects of human disturbance are usually specific for each stream type. The effects of channel bed alteration (scouring, straightening and artificial bed fixation) are completely different in lowland streams in comparison to mountain streams and even between different types of lowland streams. Thus, a stream-type specific assessment approach should always be followed.

We define a stream type as 'an artificially delineated but potentially ecologically meaningful entity with limited internal biotic (taxa composition) and abiotic (chemical and hydromorphological) variation and a biotic and abiotic discontinuity toward other types'. Stream types might serve as 'entities', for which assessment systems are applicable. A stream type should always be defined on the basis of natural or near-natural reference conditions.

It was impossible to cover all stream types occurring in Europe. Therefore, 29 stream types were selected. The criteria defined by the EU Water Framework Directive (System A, WFD Annex II, Section 1.2.1) were used for a first categorisation: ecoregions (according to Illies, 1978), size classes (based on

catchment area), geology of the catchment (siliceous, calcareous or organic formations) and altitude classes. Almost all the types had a catchment area $< 1000 \text{ km}^2$ (defined as 'small' or 'mid-sized' streams, Table 1).

In regions, where stream types are better known or regional typologies exist, additional criteria were used to further refine definition and description of types (e.g., The Netherlands, Austria and Germany). Most criteria used for this refinement are considered in System B of the EU Water Framework Directive (Annex II, Section 1.2.1) e.g. substratum, flow type and detailed geological categories.

For most stream types the major degradation factor currently affecting the stream was selected. Examples are acid stress (e.g. Northern Sweden), degradation in stream morphology (e.g. Central Europe) or organic pollution (e.g. Southern Europe). Further details are incorporated in Table 1. In some cases a combined set of stressors was assessed either individually or as a whole ('general degradation'). The rationale behind the approach was to create a stream type specific system composed of modules for stressors, which delivers not only an assessment result but simultaneously gives a clue to the cause of degradation.

Site selection

For each stream type we selected sites covering the whole range from 'reference sites' to degraded sites. Reference sites were selected using the criteria specified in Hering et al. (2003), mainly the absence of all potential stress factors, not only the absence of organic pollution. For the degraded sites a pre-classification procedure was used. After the completion of field investigations the pre-classification was refined with a set of chemical or morphological parameters to get a rough idea of the sites degree of degradation.

All steps of stream assessment must be based on near-natural reference conditions. For applied purposes, it is most important to clearly distinguish between degradation classes 'good' and 'moderate', since it is the goal of future water policy to achieve at least 'good status' for all water bodies. Therefore, the focus of AQEM was on sites that were likely to represent the classes 'high' to 'moderate'. At least 11 sampling sites per stream type were investigated, three of which belonged to each of the classes 'high' to 'moderate' and one site belonging to each of the classes 'poor' and 'bad'. Eleven was the minimum number of sites per stream type; although, in many

Table 1. Overview of stream types investigated in AQEM. Column 'ecoregion': numbered according to Illies (1978). Column 'geology class': cal = calcareous, sil = siliceous, org = organic, alluv = alluvial deposits. Column 'major degradation factors': M = degradation in stream morphology, O = Organic pollution, A = acidification, G = general degradation (not specified). Stream type I01: presently no assessment metrics are specified. Sampling seasons for the stream type in the AQEM project: sp = spring, su = summer, au = autumn, wi = winter. Number of samples: samples taken in the AQEM project

Stream type	Size class	Altitude class (m.a.s.l.)	Ecoregion	Geology class	Sampling seasons	Number of samples	Major degradation factors
A01 Mid-sized streams in the Hungarian Plains	> 100–1000 km ²	200–800	I1	sil (moraines)	su, wi	24	O
A02 Mid-sized calcareous pre-alpine streams	> 100–1000 km ²	200–800	4	cal	sp, su	26	M, O
A03 Small non-glaciated crystalline alpine streams	10–100 km ²	> 800	4	sil	sp, su	26	M, O
A04 Mid-sized streams in the Bohemian Massif	> 100–1000 km ²	200–800	9	sil	sp, su	24	M, O
C01 Mid-sized streams in the central sub-alpine mountains	> 100–1000 km ²	200–500	9	sil	sp, su	24	O
C02 Small streams in the Carpathian	10–100 km ²	200–500	10	flysch	sp, su	28	O
C03 Mid-sized streams in the Carpathian	> 100–1000 km ²	200–500	10	flysch	sp, su	22	O
D01 Small sand bottom streams in the German lowlands	10–100 km ²	< 200	14	sil	sp, su	24	M, O
D02 Organic type brooks in the German lowlands	10–100 km ²	< 200	14	org	sp	15	M, O
D03 Mid-sized sand bottom streams in the German lowlands	> 100–1000 km ²	< 200	14	sil	sp, su, au	54	M, O
D04 Small streams in lower mountainous areas of Central Europe	10–100 km ²	200–800	9	sil	sp, su	58	M, O
D05 Mid-sized streams in lower mountainous areas of Central Europe	> 100–1000 km ²	200–800	9	sil	sp, su	40	M, O
H01 Mid-altitude mid-sized siliceous streams in Northern Greece	> 100–1000 km ²	200–800	6	sil	su, wi	22	O
H02 Mid-altitude large siliceous streams in Central and Northern Greece	> 1000–10000 km ²	200–800	6	sil	su, wi	24	O
H03 Mid-altitude mid-sized calcareous streams in Western Greece	> 100–1000 km ²	200–800	6	cal	su, wi	24	O

Continued on p. 5

Table 1. Continued

Stream type	Size class	Altitude class (m.a.s.l.)	Ecoregion	Geology class	Sampling seasons	Number of samples	Major degradation factors
I01 Small-sized streams in the southern silicate Alps	10–100 km ²	>800	4	sil	sp, su, wi	33	M
I02 Small-sized, calcareous streams in the Southern Apennines	10–100 km ²	200–800	3	cal	sp, su, au	33	G
I03 Mid-sized calcareous streams in the Northern Apennines	>100–1000 km ²	200–800	3	cal	su, au	22	M
I04 Small lowland streams of the Po valley	10–100 km ²	<200	3	sil	sp, su, au	33	G
N01 Small Dutch lowland streams	≤10–100 km ²	<200	13, 14	sil	sp, su/au	142	G
N02 Small Dutch hill streams	≤10–100 km ²	<200	14	sil	sp, su/au	14	G
P01 Small-sized siliceous streams in lower mountainous areas of Southern Portugal	10–100 km ²	200–800	1	sil	sp	15	O
P02 Small-sized siliceous lowland streams of Southern Portugal	10–100 km ²	<200	1	sil	sp	11	O
P03 Medium-sized siliceous lowland streams of Southern Portugal	>100–1000 km ²	<200	1	sil	sp	13	O
S01 Small lowland streams in Northern Sweden	10–100 km ²	<200	22	sil	sp, au	30	A
S02 Small mid-altitude streams in Northern Sweden	10–100 km ²	200–500	22	sil	sp, au	30	A
S03 Small mid-altitude streams in the Boreal Highlands	10–100 km ²	200–600	20	sil	sp, au	30	A
S04 Small high-altitude streams in the Boreal Highlands	10–100 km ²	>600	20	sil	sp, au	30	A
S05 Medium-sized lowland streams in the South Swedish lowlands	100–1000 km ²	<200	14	sil	sp, au	30	A, O

cases, the number of sampling sites was up to 50 per stream type.

Each site was sampled in at least two different seasons to identify the season best suited for assessment. Most stream types have been sampled in the spring 2000, the second sampling season was different among stream types and was selected according to the hydrological features as well as experiences on seasonal variation in community composition (Table 1). The sampling resulted in altogether 901 macroinvertebrate samples, which were available for the development of stream type specific assessment methods.

Benthic macroinvertebrate sampling and identification

The sampling method applied in AQEM is based on the Rapid Bioassessment Protocols (Barbour et al., 1999), the procedures of the Environment Agency (Environment Agency 1999), the Austrian Guidelines 'Saprobiology' (Moog et al., 1999) and ISO 7828 (1985). These guidelines have been combined by the AQEM partners into a standard procedure for collecting and analysing macroinvertebrate samples within the AQEM project (AQEM consortium, 2002).

The AQEM sampling method focuses on a multi-habitat scheme designed for sampling major habitats in proportion to their presence within a sampling reach. A sample consists of 20 'sampling units' taken from all microhabitat types at the sampling site with a share of at least 5% coverage. A 'sampling unit' is a sampling accomplished by positioning the net and disturbing the substratum ('kicking', 'sweeping') for an area that equals the square of the frame width upstream of the net (0.25×0.25 m) or by pushing the net over a length of 0.25 m of the habitat. Alternatively, an open Surber sampler has sometimes been used instead of a pond-net. The 20 'sampling units' must be distributed according to the share of microhabitats (Fig. 1; Table 2). This procedure results in sampling approximately 1.25 m^2 stream bottom area.

More specifically, the 'multi-habitat sampling' procedure was performed in the following steps:

1. Before sampling, the site protocol (field sheets describing site characteristics) was completed. However, the sampling area must not be disturbed by physical contact, if at all possible. Therefore, after sampling, the site protocol information was reviewed for accuracy and completeness.

2. The coverage of all microhabitats (Table 2) with at least 5% cover was recorded to the nearest 5% interval on a standard field pro-forma. The presence of each other microhabitat (<5% cover) was indicated by the symbol '+' on the same form.
3. Based on the estimation of microhabitat coverage the number of sampling units in the individual habitats was determined and indicated in the site protocol.
4. The 20 'sampling units' were sampled starting at the downstream end of the reach and proceeding upstream.
5. After every three 'sampling units' (or more frequently if necessary) the material collected was rinsed through the net two to three times, using clean running stream water.
6. If riffle and pool areas were clearly distinguishable in the stream stretch investigated, the current velocity, the habitat conditions at the microhabitat scale and faunal assemblages could be expected to be different. Accordingly, for some stream types, the 'sampling units' collected from riffles and pools were stored and treated separately. Usually, different numbers of 'sampling units' were taken in riffles and pools (e.g. 13 sampling units in riffles and 7 in pools) – depending on their relative importance (area within the sampling site).
7. Large pieces of wood and stones were removed after being rinsed and inspected for clinging or sessile organisms. For some South European stream types the samples were fully sorted in the field.
8. Large and rare organisms, which could easily be identified in the field (such as large mussels), were noted on the site protocol form, removed from the sample and returned to the stream.
9. The complete sample was sieved through a coarse meshed sieve ($1000 \mu\text{m}$ for sand bottom streams; $2000 \mu\text{m}$ for stony bottom streams), producing a coarse and a fine fraction. Both were usually sorted separately and completely in the laboratory.

This procedure resulted in up to four fractions of a sample: coarse and fine fraction (larger and smaller than 1000 or 2000μ , respectively) for riffle and pool fractions. Taxa identification was performed to the best attainable level. While most taxonomic groups have been identified to species level in Central and Northern Europe, genus and family level were usually applied for Southern European streams due to lack of taxonomic knowledge and identification keys.

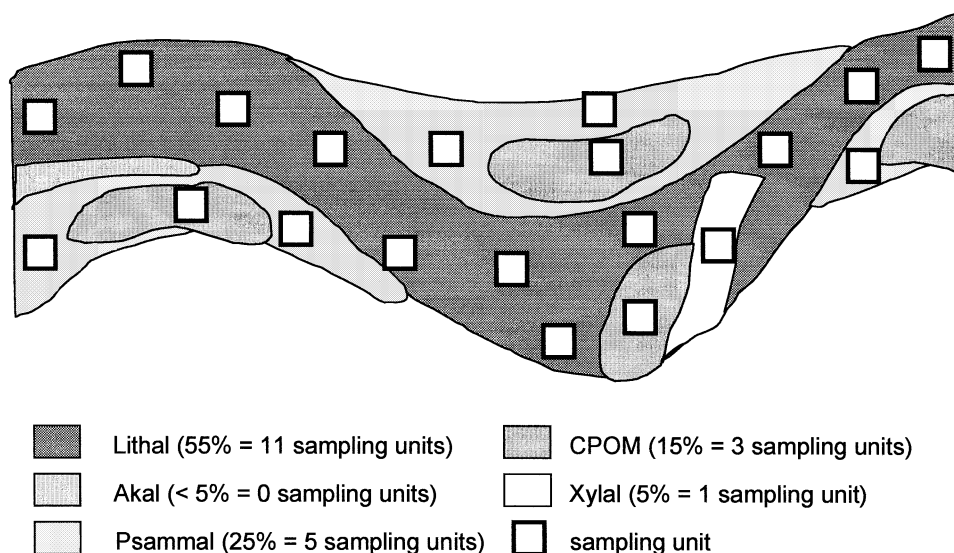


Figure 1. Example of 'sampling unit' position in a theoretical sampling site according to the 'multi habitat sampling' method applied in AQEM.

Table 2. Defined list of microhabitats for use with the AQEM site protocol and selecting numbers and locations of AQEM sampling units. For mineral substrates, the largest occurring grain-size is decisive.

Mineral substrates	Definition
Hygropetric sites	water layer on mineral substrata
Megalithal (>40 cm)	upper sides of large cobbles, boulders and blocks, bedrock
Macrolithal (>20 cm to 40 cm)	coarse blocks, head-sized cobbles, with a variable percentages of cobble, gravel and sand
Mesolithal (>6 cm to 20 cm)	fist to hand-sized cobbles with a variable percentage of gravel and sand
Microlithal (>2 cm to 6 cm)	coarse gravel, (size of a pigeon egg to child's fist) with variable percentages of medium to fine gravel
Akal (>0.2 cm to 2 cm)	fine to medium-sized gravel
Psammal/psammopelal (>6 μ m to 2 mm)	sand and mud
Argyllal (<6 μ m)	silt, loam, clay (inorganic)
Biotic microhabitats	Definition
Phytal	floating stands or mats, lawns of bacteria or fungi, and tufts, often with aggregations of detritus, moss or algal mats (interphytal: habitat within a vegetation stand, plant mats or clumps)
Algae	filamentous algae, algal tufts
Submerged macrophytes	macrophytes, including moss and Characeae
Emergent macrophytes	e.g. <i>Typha</i> , <i>Carex</i> , <i>Phragmites</i>
Living parts of terrestrial plants	fine roots, floating riparian vegetation
Xylal (wood)	tree trunks, dead wood, branches, roots
CPOM	deposits of coarse particulate organic matter, e.g. fallen leaves
FPOM	deposits of fine particulate organic matter
Sewage bacteria and -fungi and saprobel	sewage bacteria and -fungi, (<i>Sphaerotilus</i> , <i>Leptomitius</i>), sulfur bacteria (e.g. <i>Beggiatoa</i> , <i>Thiothrix</i>), sludge
Organic mud	mud and sludge (organic) = pelal
Debris	organic and inorganic matter deposited within the splash zone area by wave motion and changing water levels, e.g. mussel shells, snail shells

'Site protocol' and development of a stream-type specific classification

In addition to the data on macroinvertebrate community composition, a large number of parameters describing the streams and their catchments were recorded for all sampling sites, using a harmonised 'site protocol'. The site protocol comprises more than 200 parameters and covers:

- features of the catchment, such as geology and land-use (to be recorded from maps);
- characters of an upstream and downstream section (5 km upstream and downstream of sampling point in case of small streams with a catchment area of <100 km², 10 km upstream and downstream of sampling point in case of medium-sized streams with a catchment area of 100–1000 km² and 50 km upstream and downstream of sampling point in case of small streams with a catchment area of <100 km²), such as transverse structures and residual flow sections;
- characters of the 'sampling site' (250 m upstream and downstream of sampling point in case of small streams with a catchment area of <100 km², 500 m upstream and downstream of sampling point in case of medium-sized streams with a catchment area of 100–1000 km² and 1000 m upstream and downstream of sampling point in case of small streams with a catchment area of <100 km²), such as microhabitat composition (which is further used to determine the location of sampling units, see above), chemical parameters, discharge and flood-plain characteristics.

The site protocol data were used for both multivariate analyses and for deriving a rough classification of the site's stage of degradation. The latter was used to replace the pre-classification and, thus, to provide a more precise measure of stress. This resulted either in a classification of the sites into the five quality classes based on abiotic characteristics such as local morphology and chemistry or alternatively in a dimensionless value of each site describing its state of degradation (Buffagni et al., 2004; Lorenz et al., 2004).

Testing and selection of metrics

A large number of metrics (Table 3) was tested independently for each of the 29 stream types in order to the extent of their correlation with the degradation of a site, as determined by assessment of the site protocol data. A number of metrics, such as feeding types and

microhabitat or current velocity preferences, require scores for the taxa. Therefore, a taxa list was generated that included autecological information for the individual taxa. In order of prioritisation, the autecological information was taken from:

- Moog (1995)
- Schmedtje & Colling (1996)
- Verdonschot (1990); van den Hoek & Verdonschot (1994)
- Information collected by the AQEM consortium.

The resulting taxa list was the basis of all metric calculations and is further described in Section 4.1. Basically we followed the 'kingdom of metrics – approach' concerning Karr & Chu (1999). Only metrics capable of distinguishing between one or more 'stress' classes are suited for assessment purposes. Metrics that clearly respond to specific pollutants or stressors are most useful as a diagnostic tool. Furthermore, the metrics used should in combination cover diverse aspects of structure, composition, health and function of the aquatic biota. When considering these general guidelines, it becomes obvious that different metrics are required for assessing different stream types.

The process of selecting the metrics to be used for the assessment was different for each of the stream types. In most cases correlation analyses between the results of a metric and the abiotic classification were performed (Ofenböck et al., 2004) to identify the metrics that showed a quantitative dose-response change across the whole range of disturbance. In some cases, however, the classification of a site was solely based on the faunal assemblages and no abiotic parameters were taken into account (Vlek et al., 2004).

Further criteria for the selection of suitable assessment metrics were that the selected metrics should deliver information on different components of the community and that they should be consistent with the country's traditions in stream monitoring. The selection process resulted in up to 18 suitable core metrics for the individual AQEM stream types. In exceptional cases only one metric was used (usually if a single metric was extremely well correlated to a specific stressor). A complete list of metrics used for assessing the individual stream types and stressors is given in Table 4. The selected metrics were finally combined into multimetric indices.

Table 3. Metrics calculated by the AQEM River Assessment Program (version 1.0). In addition, a number of combinations of different metrics is calculated. Criteria of the Water Framework Directive met: tx = taxonomic composition; ab = abundance; ra = ratio sensitive/insensitive taxa; di = diversity. Stressors detected: po = organic pollution; mo = degradation in stream morphology; ac = acid stress; ge = general degradation.

Metric	Criteria of the WFD met	Stressors detected	Reference
Abundance (ind m ⁻²)	ab	po, mo, ge	Zelinka & Marvan (1961)
Number of Taxa	tx, di	po, mo, ac, ge	
Saprobic Index (Zelinka & Marvan)	ra	po	
Saprobic valencies			
• Xenosaprobic (%) • Oligosaprobic (%) • Beta-mesosaprobic (%) • Alpha-mesosaprobic (%) • Polysaprobic (%)			
German Saprobic Index (old version)	ra	po	DEV (1992)
• Measure of dispersion ('Streuungsmaß') • Measure of Abundance ('Abundanzziffer') • Number of Indicator Taxa • Water Quality Class			
German Saprobic Index (new version)	ra	po	unpublished, standardisation in progress
• Measure of dispersion ('Streuungsmaß') • Measure of Abundance ('Abundanzziffer') • Number of Indicator Taxa	ra	po	
– Water Quality Class			
Dutch Saprobic Index	ra	po	Verdonschot (1990)
Czech Saprobic Index	ra	po	Czech Standard No. 757716
Biological Monitoring Working Party (BMWP)	ra	po, mo, ge	Armitage et al. (1983)
Average score per Taxon	ra	po	Armitage et al. (1983)
BMWP (Spanish version)	ra	po	Alba-Tercedor & Sanchez-Ortega (1988)
Danish Strema Fauna Index (DSFI)	tx, ra, di	po, ge	Skriver et al. (2001)
• DSFI Diversity Groups			
Belgian Biotic Index (BBI)	tx, ra	po	De Pauw et al. (1992)
Indice Biotico Estesio (IBE)	tx, ra, di	po	Ghetti (1997)
• IBE Quality Class • IBE Systematic Units			
Altered Indice Biotico Estesio (IBE AQEM)	tx, ra, di	po	
• IBE AQEM Quality Class			
Mayfly Average Score (MAS)	tx, ra, di	po, mo, ge	Buffagni (1997, 1999)
• MAS Integrity Class • MAS Operational Units • Mayfly Total Score (MTS)			
Mayfly Average Score (MAS) (Large Rivers)	tx, ra, di	po, mo, ge	
• MAS Large Rivers Integrity Class • MAS Large Rivers Operational Units • Mayfly Total Score (MTS)			
Diversity (Simpson-Index)	di	po, mo, ge	Simpson (1949)
Diversity (Shannon-Wiener-Index)	di	po, mo, ge	Shannon & Weaver (1949)
Diversity (Margalef Index)	di	po, mo, ge	Margalef (1984)
Evenness	di	po, mo, ge	

Continued on p. 10

Table 3. Continued

Metric	Criteria of the WFD met	Stressors detected	Reference
Acid Class (according to Braukmann)	ra	ac	Braukmann (2000)
• Share acid class 1 (no acidification) • Share acid class 2 (periodical slight acidification) • Share acid class 3 (periodical serious acidification) • Share acid class 4 (permanent acidification)			
Acid Index (Hendrikson & Medin)	ra	ac	Hendrikson & Medin (1986)
German Fauna Index	ab, ra	mo	Lorenz et al. (2004)
• German Fauna Index D01 • German Fauna Index D02 • German Fauna Index D03 • German Fauna Index D04 • German Fauna Index D05			
Portuguese Index	tx, ra	po, ge	Pinto et al. (2004)
Number of sensitive taxa (Austria)	tx, ra	po, ge	Moog et al. (1999)
Zonation (percentage of community preferring a certain zone)	tx, ra, di	po, mo	Moog et al. (1995); Schmedtje & Colling (1996)
• Crenal (spring) (%) • Hypocrenal (spring-brook) (%) • Epirhithral (upper-trout region) (%) • Metarhithral (lower-trout region) (%) • Hyporhithral (greyling region) (%) • Epipotamal (barbel region) (%) • Metapotamal (brass region) (%) • Hypopotamal (brackish water) (%) • Littoral (%) • Profundal (%)			
Current preference (percentage of community preferring a certain current velocity)	tx, ra	mo	Schmedtje & Colling (1996)
• Type LB (limnobiont, occurring only in standing waters) (%) • Type LP (limnophil, preferably occurring in standing waters; avoids current; rarely found in slowly flowing streams) (%) • Type LR (limno- to rheophil, preferably occurring in standing waters but regularly occurring in slowly flowing streams) (%) • Type RL (rheo- to limnophil, usually found in streams; prefers slowly flowing streams and lentic zones; also found in standing waters) (%) • Type RP (rheophil, occurring in streams; prefers zones with moderate to high current) (%) • Type RB (rheobiont, occurring in streams; bound to zones with high current) (%) • Type IN (indifferent, no preference for a certain current velocity) (%)			

Continued on p. 11

Table 3. Continued

Metric	Criteria of the WFD met	Stressors detected	Reference
Microhabitat preference (percentage of community preferring a certain microhabitat)	tx, ra, di	po, mo	Schmedtje & Colling (1996)
• Type Pel (Pelal: mud; grain size <0.063 mm) (%) • Type Arg (Argyllal: silt, loam, clay; grain size <0.063 mm) (%) • Type Psa (Psammal: sand; grain size 0.063–2 mm) (%) • Type Aka (Akal: fine to medium-sized gravel; grain size 0.2–2 cm) (%) • Type Lit (Lithal: coarse gravel, stones, boulders; grain size >2 cm) (%) • Type Phy (Phytal: algae, mosses and macrophytes including living parts of ter-restrial plants) (%) • Type POM (particulate organic matter, such as woody debris, CPOM, FPOM) (%) • Type Oth (other habitats) (%)			
Feeding types (percentage of community)	tx, ra, di	po, mo, ge	Moog et al. (1995); Schmedtje & Colling (1996)
• Grazers and scrapers (%) • Miners (%) • Xylophagous Taxa (%) • Shredders (%) • Gatherers/Collectors (%) • Active filter feeders (%) • Passive filter feeders (%) • Predators (%) • Parasites (%)			
RETI (Rhithron Feeding Type Index)			Schweder (1992), Podraza et al. (2000)
Locomotion type (percentage of community)	tx, ab, ra	po, mo, ac	Schmedtje & Colling (1996)
• Swimming/skating (%) • Swimming/diving (%) • Burrowing/boring (%) • Sprawling/walking (%) • (Semi)sessil (%) • Others (e.g. climbing) (%)			
Taxonomic group (%) (calculated for 30 groups)	tx, ra	po, mo, ge	
Taxonomic group (number of taxa) (calculated for 32 groups)	tx, di	po, mo, ge	
Taxonomic group (abundance) (calculated for 32 groups)	tx, ab, di	po, ge	
Number of Families	di	po, mo, ge	
Number of Genera	di	po, mo, ge	
Index of Biocoenotic Region	tx, ra	po, mo	Moog & Ofenböck, in prep.

Table 4. Metrics tested with the faunal dataset for the individual stream types. For metric abbreviations and literature references compare Table 3. Metric selection according to AQEM River Assessment Program, version 1.0. Deviations according to the Greek and Portuguese stream types are described in Pinto et al. (2004) and Skoulikidis et al. (2004)

AQEM code	Stream type	Metrics used to assess the ecological quality	Predicted response to increasing perturbation
A01	Mid-sized streams in the Hungarian Plains	organic pollution	
		• Saprobic Index Zelinka & Marvan	increase
		• Number of Ephemeroptera+Plecoptera-taxa	decrease
		• (%) Ephemeroptera+Plecoptera-taxa/total taxa (sp)	decrease
		• (%) Ephemeroptera+Plecoptera individuals / total individuals	decrease
		• Total number of families	decrease
		• Number of sensitive taxa	decrease
		• (%) Littoral+profundal	increase
		• Abundance of Plecoptera	decrease
		• (%) Shredder	decrease
		• Diversity (Margalef)	decrease
A02	Mid-sized calcareous pre-alpine streams	organic pollution	
		• Saprobic Index Zelinka & Marvan	increase
		degradation in stream morphology	
		• Number of EPT-taxa	decrease
		• Total number of taxa	decrease
		• (%) EPT-taxa/total taxa	decrease
		• Number of sensitive taxa	decrease
		• Abundance of Plecoptera	decrease
		• Abundance of Trichoptera	decrease
		• Diversity (Margalef)	decrease
A03	Small non-glaciated crystalline alpine streams	organic pollution	
		• Saprobic Index Zelinka & Marvan	increase
		degradation in stream morphology	
		• Number of EPT-taxa	decrease
		• Total number of taxa	decrease
		• Number of sensitive taxa	decrease
		• Abundance of Plecoptera	decrease
		• Ratio Oligochaeta and Diptera/total-taxa	increase
		• Abundance of Oligochaeta	increase
		• RETI	decrease
		• Diversity (Margalef)	decrease
		• (%) Littoral and Profundal preferences	increase
A04	Mid-sized streams in the Bohemian Massif	organic pollution	
		• Saprobic Index Zelinka & Marvan	increase
		degradation in stream morphology	
		• Number of EPT-taxa	decrease
		• Abundance of all taxa	variable
		• Index of Biocenotic Region	variable
		• (%) Oligochaeta and Diptera taxa	increase
		• (%) Littoral preferences	increase
		• (%) Gatherers/collectors	increase
		• Total number of taxa	decrease
		• Abundance of Trichoptera	decrease

Continued on p. 13

Table 4. Continued

AQEM code	Stream type	Metrics used to assess the ecological quality	Predicted response to increasing perturbation
C01	Mid-sized streams central sub-alpine mountains	organic pollution • Czech saprobic index • ASPT • RETI	increase decrease decrease
C02	Small streams in the Carpathian	organic pollution • Czech saprobic index • Number of Plecoptera taxa • Number of Ephemeroptera taxa	increase decrease decrease
C03	Mid-sized streams in the Carpathian	organic pollution • Czech saprobic index • Number of EPT taxa	increase decrease
D01	Small sand bottom stream in the German lowlands	organic pollution • German saprobic index (new version) degradation in stream morphology • German Faunaindex D01 • (%) Plecoptera • (%) Rheophilous preferences • (%) Gatherers/collectors (ind.) • (%) Littoral preferences • (%) Pelal preferences	increase decrease decrease decrease increase increase increase
D02	Organic type brook in the German lowlands	organic pollution • German saprobic index (new version) degradation in stream morphology • German Faunaindex D02	increase decrease
D03	Mid-sized sand bottom stream in the German lowlands	organic pollution • German saprobic index (new version) degradation in stream morphology • German Faunaindex D03 • (%) Trichoptera • (%) Rheophilous preferences • (%) Gatherers/collectors • (%) Littoral preferences • (%) Pelal preferences	increase decrease decrease decrease increase increase increase
D04	Small streams in lower mountainous areas of Central Europe	organic pollution • German saprobic index (new version) degradation in stream morphology • German Faunaindex D04 • BMWP • Shannon–Wiener Diversity • (%) Hyporhithral preferences • (%) Hypocrenal preferences • (%) Akal preferences • (%) Phytal preferences	increase decrease decrease decrease increase decrease decrease increase

Continued on p. 14

Table 4. Continued

AQEM code	Stream type	Metrics used to assess the ecological quality	Predicted response to increasing perturbation
D05	Mid-sized streams in lower mountainous areas of Central Europe	organic pollution • German saprobic index (new version) degradation in stream morphology • German Faunaindex D05 • Number of EPTCBO taxa • (%) Xylophagous taxa + (%) shredder + (%) active filter feeders + (%) passive filter feeders • (%) Akal + (%) lithal + (%) psammal • Shannon–Wiener-Diversity	increase decrease decrease decrease decrease decrease
H01	Mid-altitude mid-sized siliceous streams in North-Eastern Greece	organic pollution • Type LR • Type RP • (%) EPT-taxa • BMWP	increase decrease decrease decrease
H02	Mid-altitude large siliceous streams in Central and Northern Greece	organic pollution • ASPT • BMWP • DSFI • Diversity Groups • IBE • Simpson Diversity • (%) Littoral preferences • (%) Predators • Number of EPT – taxa • Type RP • (%) Metapotamal preferences • (%) EPT - taxa	decrease decrease decrease decrease decrease decrease decrease decrease decrease decrease
H03	Mid-altitude mid-sized calcareous streams in Western Greece	organic pollution • German Saprobic Index (old version) • (%) Hypopotamal preferences • (%) Parasites% • (%) Profundal • (%) Passive filter feeders	increase increase increase increase increase
I02	Small-sized, calcareous streams in the Southern Apennines	general degradation • ASPT • BMWP • MTS (Mayfly total score) • Number of Plecoptera taxa • Number of Trichoptera taxa • Number of MAS Operational Units • TROPHIC_Sel_Grazers • Abundance of <i>Alainites muticus</i> + <i>Nigrobaetis di-</i> <i>gitatus</i>	decrease decrease decrease decrease decrease decrease decrease decrease

Continued on p. 15

Table 4. Continued

AQEM code	Stream type	Metrics used to assess the ecological quality	Predicted response to increasing perturbation
		• Abundance of Leptophlebiidae • Selected Ephemeroptera_GS • Abundance of <i>Cordulegaster</i> and <i>Dinocras</i> • Abundance of <i>Amphinemura</i> and <i>Protonemura</i> • Selected Trichoptera_GS • DIPTERA_Good_G • DIPTERA_Bad_SIPH_G	decrease decrease decrease decrease decrease decrease increase
I03	Mid-sized calcareous streams in the Northern Appennines	degradation in stream morphology • BMWP • MTS (Mayfly Total Score) • Number of Plecoptera and Trichoptera taxa • Number of MAS Operational Units • (%) Argyllal preferences • (%) Filter feeders • (%) Borrowing locomotion types • Abundance of selected Ephemeroptera • Abundance of selected Plecoptera • Abundance of selected non EP taxa • Abundance of <i>Dugesia</i> and <i>Lymnaea</i> • Abundance of all taxa / abundance of Diptera taxa	decrease decrease decrease decrease decrease increase decrease decrease decrease decrease increase decrease
I04	Small streams in the lowlands of the Po valley	general degradation • ASPT • BMWP • MTS (Mayfly Total Score) • Number of Plecoptera and Trichoptera taxa • Diversity Groups (DSFI) • Number of MAS Operational Units • Abundance of Sel_Ephemeroptera_GN • Abundance of Sel_Trichoptera_GN • Abundance of <i>Leuctra</i> and <i>Calopteryx</i> • Abundance of Elmidae • Abundance of Lumbricidae • Abundance of Tubificidae	decrease decrease decrease decrease decrease decrease decrease decrease decrease decrease decrease increase
N01	Small Dutch lowland streams	general degradation • Saprobic Index Zelinka & Marvan • (%) Hypopotamal preferences • (%) Type PEL • (%) Type RP • (%) Hypopotamal-(%) EPT/(%) Oligochaeta • (%) Gastropoda-(%) EPT/(%) Oligochaeta • (%) EPT/(%) Oligochaeta • Number of EPT taxa/number of Oligochaeta taxa	Combination of metrics results in the Quality Class. Most individual metrics do not respond in a linear way to perturbation.

Continued on p. 16

Table 4. Continued

AQEM code	Stream type	Metrics used to assess the ecological quality	Predicted response to increasing perturbation
		• (%) (Grazers + scrapers)/ (%) (gatherers/collectors + filter feeders) • (%) Gastropoda	
N02	Small Dutch hill streams	general degradation • Saprobic Index (Zelinka & Marvan) • (%) Hypopotamal preferences • Total number of taxa • (%) Passive filter feeders • (%) EPT/(%) Oligochaeta • Number of EPT taxa/number of Oligochaeta taxa • (%) Trichoptera • (%) EPT/(%) Oligochaeta - (%) Gastropoda • (%) EPT/(%) Oligochaeta - (%) type RP • (%) EPT/(%) Oligochaeta - (%) type PEL	Combination of metrics results in the Quality Class. Most individual metrics do not respond in a linear way to perturbation.
P01	Small-sized siliceous streams in lower mountainous areas of Portugal	organic pollution • Portuguese Index	decrease
P02	Small-sized siliceous lowland streams of Portugal	organic pollution • Portuguese Index	decrease
P03	Medium-sized siliceous lowland streams of Portugal	organic pollution • Portuguese Index	decrease
S01	Small lowland streams in Northern Sweden	acidification • Acid index (Henrikson & Medin) • Number of EPT taxa • (%) Swimming/diving taxa	decrease decrease decrease
S02	Small mid-altitude streams in Northern Sweden	acidification • Acid index (Henrikson & Medin) • Number of EPT taxa • (%) Grazers/scrapers	decrease decrease decrease
S03	Small mid-altitude streams in the Boreal Highlands	acidification • Acid index (Henrikson & Medin) • Number of EPT taxa	decrease decrease
S04	Small high-altitude streams in the Boreal Highlands	acidification • Acid index (Henrikson & Medin) • Number of EPT taxa	decrease decrease
S05	Medium-sized lowland streams in the South Swedish lowlands	organic pollution • ASPT • DSFI acidification • Acid index (Henrikson & Medin)	decrease decrease decrease

The AQEM assessment system

The AQEM assessment system currently covers 28 European stream types and is designed to meet the following objectives:

- to classify a sampling site into an Ecological Quality Class ranging from 5 (high) to 1 (bad) based on a macroinvertebrate taxa list, which has been ob-

tained from sampling the site using the multihabitat sampling method described in Section 2.1;

- to give information on the possible causes of degradation in order to help direct future management practices.

The stream type specific approach applied in AQEM resulted in different combinations of assessment metrics to be used for different stream types. However, the system always follows the same evaluation scheme and each stream-type specific method fits into the general assessment framework. This framework is based on:

- Stressor-specific approach: for each stream type the ‘main’ degradation factor currently affecting the stream is assessed (either acid stress, degradation in stream morphology or organic pollution). In some cases more than one stressor is assessed individually or as an overall disturbance.
- Multimetric system: for each stream type, calculation methods have been identified that best indicate a site’s state of degradation. The results of the individual calculation methods are then combined in a ‘multimetric formula’.
- The multimetric result is converted into the final score ranging from 5 (high quality) to 1 (bad quality). The classes represent a gradient from either reference or best-available conditions to degraded conditions.

The general architecture of a multimetric approach, as applied in the AQEM assessment system, is outlined in Fig. 2. It consists of the following steps:

1. The starting point is the taxa list obtained from the sampling site, which is to be assessed.
2. Based on this taxa list a number of metrics is calculated.
3. Generally, the metric results are individually converted into scores by comparing their values with the values of the same metrics in stream-type specific reference condition.
4. The scores or results of the metrics are combined in a simple multimetric index (usually the average of all scores).

This procedure enables the user to view both the final assessment result (Ecological Quality Class) and the individual metric results, allowing further interpretation of the data for future management procedures.

Whenever possible, the metrics used for calculating the ecological quality of a stream type were categorised beforehand according to their ability to

Table 5. Summary of the contents of the AQEM taxa list

Parameter	Number
Number of species included	7980
Number of higher taxa included	1577
Total number of taxa	9557
Number of insect taxa	8422
Number of non-insect taxa	1135
Number of taxa for which information on saprobic valencies is available	1946
Number of taxa for which information on feeding types is available	3264
Number of taxa for which information on preference for a certain stream zone is available	2211
Number of taxa for which information on current preferences is available	1722
Number of taxa for which information on microhabitat preferences is available	1685
Number of taxa for which information on locomotion type is available	623

detect the impact of certain stressors. This was possible for nine out of 28 AQEM stream types. In these cases step 4 of the above scheme is slightly different: The scores of those metrics, which address the same stressor, are combined in a simple multimetric index (usually the average of all scores) for a stressor-specific assessment module, e.g. a module for ‘organic pollution’ or for ‘degradation in stream morphology.’ The results of the individual modules are finally combined using the worst result of all modules as the ecological quality class. This refined procedure further eases data interpretation as results at three different scales are available: (1) the Ecological Quality Class, (2) the Quality Classes of the individual modules and (3) the results of the individual metrics. The rationale behind such ‘stressor specific approach’ is initiated in the aims of the Water Framework Directive. As soon as a sampling site is classified as being ‘moderate’ in quality the individual metric values can be used to interpret what the causes of degradation may be.

The AQEM tools

The AQEM assessment method was made available for public use in March 2002 through three tools (taxalist, software and manual), which are accessible via the project homepage (www.aqem.de).

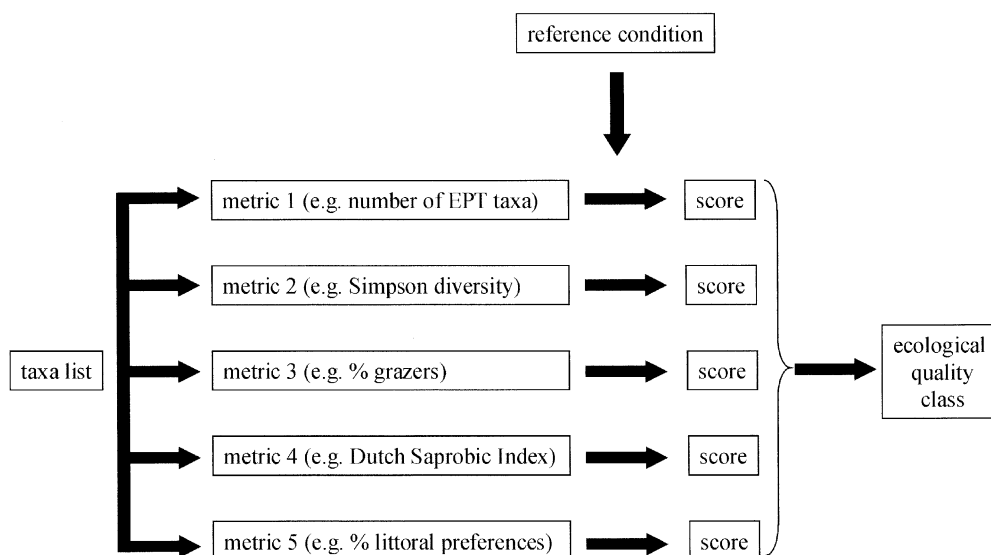


Figure 2. General scheme of a multimetric calculation.

The AQEM taxalist

The AQEM taxa list serves the following purposes:

- It contains autecological information on a number of the AQEM taxa. The taxa list is the basis for the AQEM European Stream Assessment Program, the AQEM data input program and the Austrian assessment software ECOPROF (Moog et al., 2001).
- It provides a list of species occurring in the individual European countries, particularly those that are represented in AQEM.
- It provides hierarchical taxonomic information on each species with regard to order, family, subfamily, genus, authority and date of description.
- It serves as a tool for many aspects of quality assurance.

The list was compiled by the generation of draft lists of aquatic invertebrates occurring in the eight AQEM partner countries. Additional lists were created for Slovakia and – for some taxonomic groups only – for Norway, Finland and Denmark. This was done using published checklists. Any taxa not previously recorded in a country but found in AQEM samples of that country, were added to the ‘national lists’. The resulting AQEM list was checked by taxonomic experts for correct nomenclature and taxonomy. The taxonomical precision of the list differs for individual countries; for Central and Northern Europe countries the lists should be nearly complete with respect to running waters, while for other countries considerable gaps may

remain. The final AQEM taxa list comprised 7980 species and 1577 higher taxa (Table 5). Autecological information is included for many of these.

The AQEM software

The AQEM River Assessment Program performs all calculations necessary for applying the AQEM system:

- calculation of the Ecological Quality Class of a sampling site, based on a standardised macroinvertebrate taxa list, by performing the stream-type specific calculations specified in Table 4;
- calculation of a large number of additional metrics, which give assistance for further data interpretation. The complete list of metrics covered by the program is displayed in Table 3.

The AQEM software is only a calculation program and not designed for storing data. The different European water authorities use their own databases and only need the AQEM software for data analysis. The AQEM software is based on EXCEL as common and comparatively compatible computing system, to which most databases are able to export data sheets. The AQEM software is capable of importing a taxa list in either EXCEL or ASCII file format and exporting results to either EXCEL or ASCII files. Preparing data sets for import into the AQEM software must be done using a different program software package, e.g. EXCEL or a text editor. The calculation of the metrics

is based on the AQEM taxa list, which is incorporated in the program as a separate database.

The AQEM manual

Detailed information on the application of the assessment procedure, including all stream type specific deviations, is given in a manual for the users (AQEM consortium 2002). Amongst other things, the manual includes:

- background information on stream typologies and detailed descriptions of the AQEM stream types;
- a detailed description of reference conditions;
- restrictions on its application;
- guidelines for selecting suitable sampling sites;
- a site protocol modified for the use in applied water management and a manual on its completion;
- descriptions of the sampling and sample processing methods including some variations in the procedure that are specific to certain stream types only;
- identification levels and literature;
- guidelines for processing taxalists prior to calculation (taxonomic adjustment);
- a general description of the AQEM assessment approach and detailed descriptions of all assessment metrics;
- the AQEM software manual;
- guidelines for interpretation of the AQEM results.

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